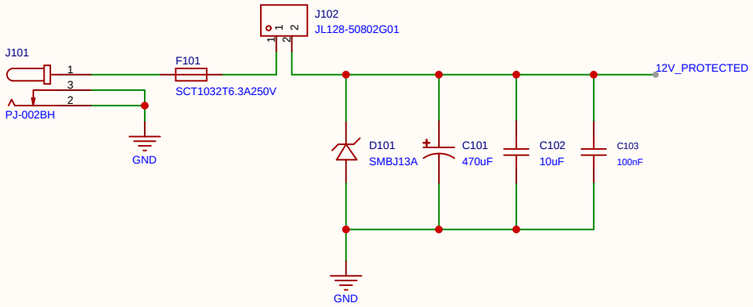
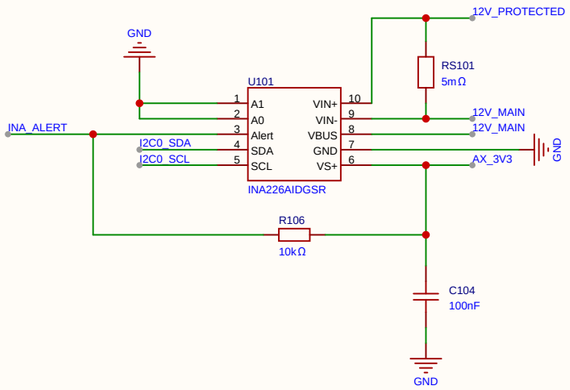
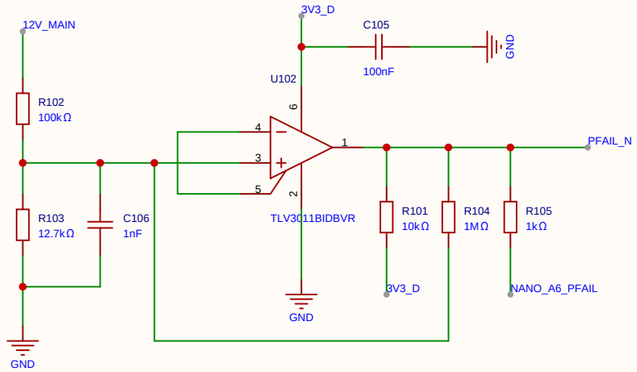


主电源入口与瞬态保护

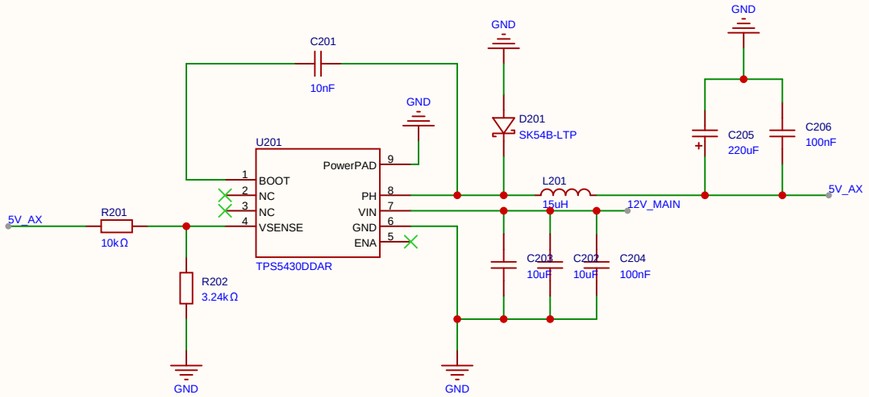


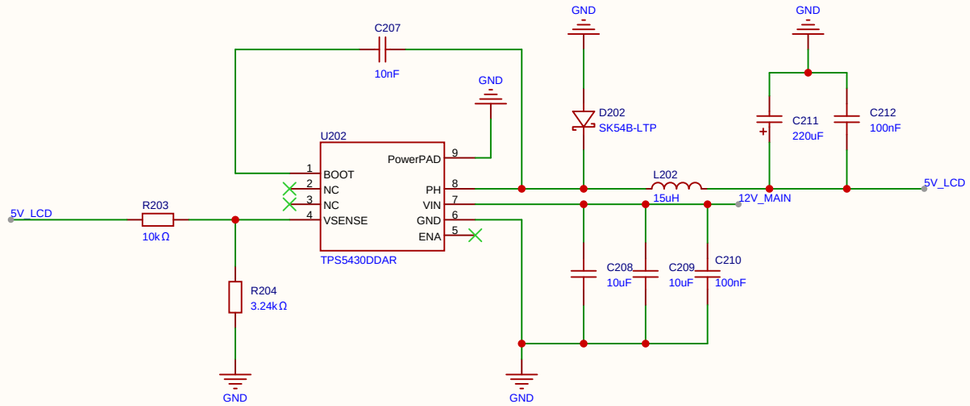
母线电流与功率监测

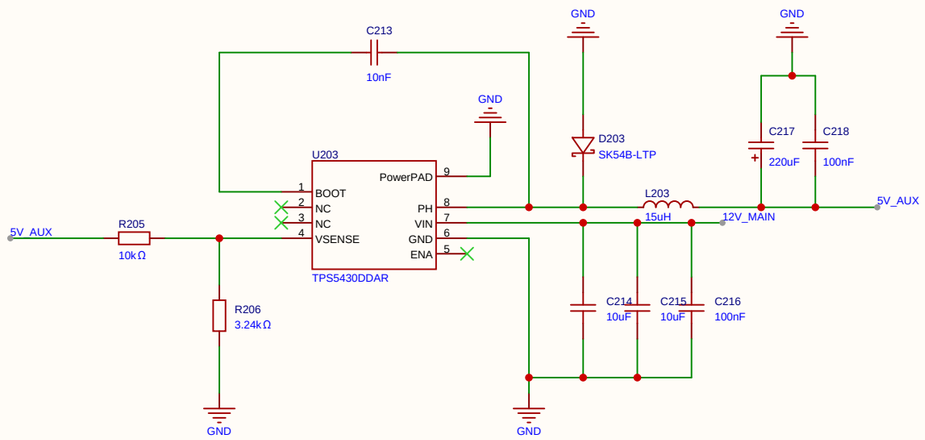


掉电硬件中断检测

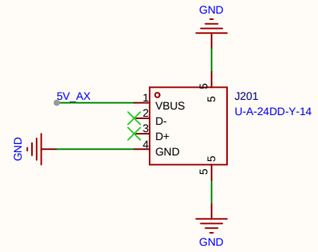
原理图	主载板原理图	创建日期	2026-10-14
		更新日期	2026-10-17
板子	主载板	图页	电源输入与监测
绘制		FPGA_2026-10-14_17-44-11	
审阅			
		版本	尺寸	页 1 共 8
嘉立创EDA		V1.0	A4	嘉立创EDA



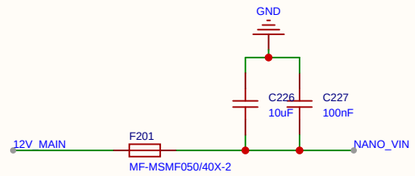




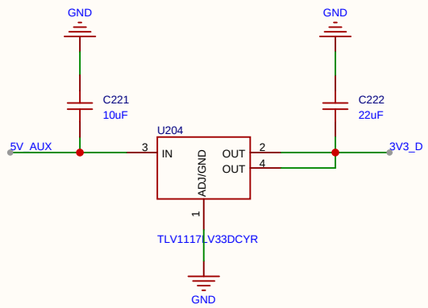
三路5V BUCK降压电路



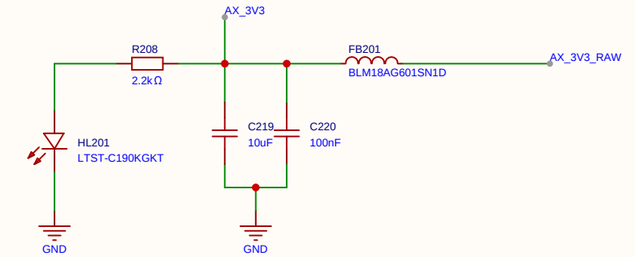
AX7010供电口



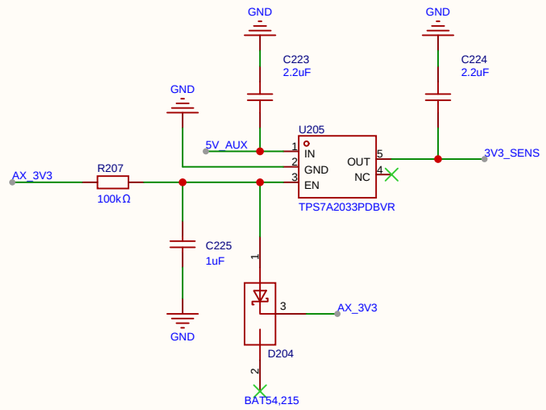
Nano ESP32供电与保护支路



数字常电 LDO 回路

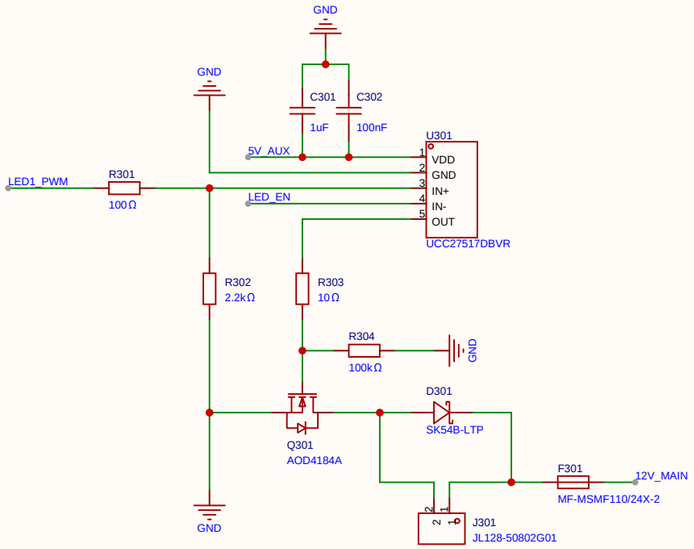


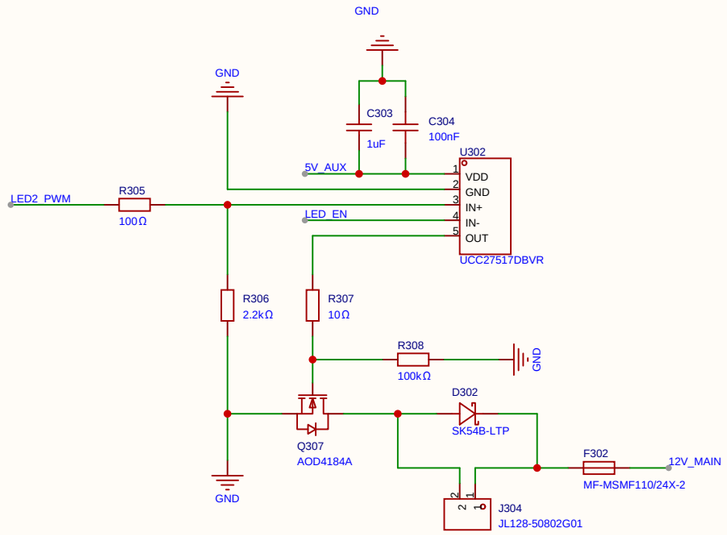
磁珠滤波与硬件指示灯



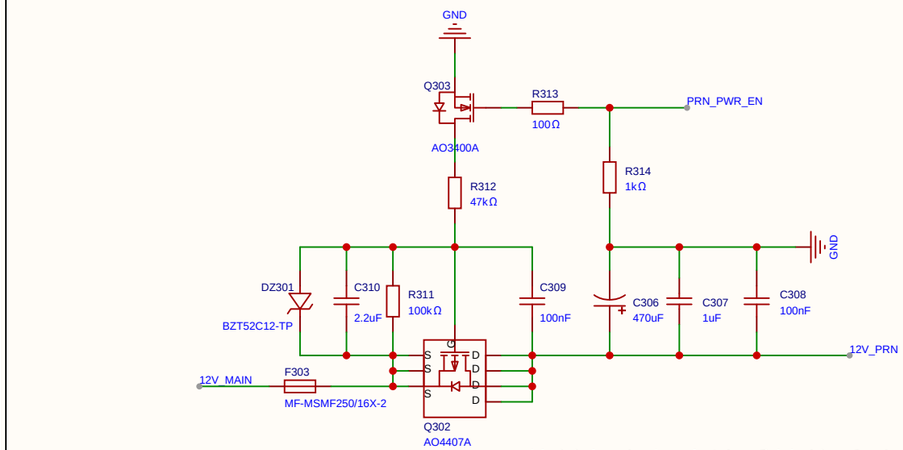
传感器精密超低噪声 LDO 与时序联动回路

原理图	主载板原理图	创建日期	2026-10-14
		更新日期	2026-10-18
板子	主载板	图页	电源转换与分配
绘制		FPGA_2026-10-14_17-44-11	
审阅			
		版本	尺寸
嘉立创EDA		V1.0	A4
		页 2 共 8	
		嘉立创EDA	

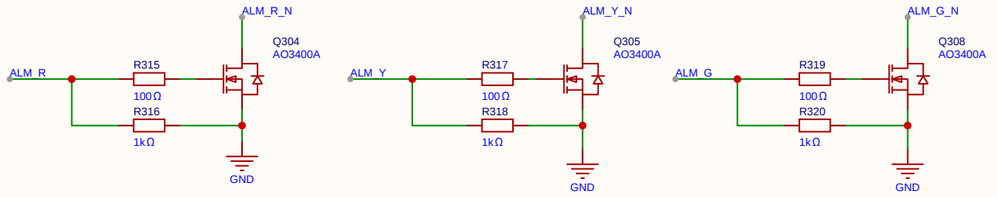




补光灯驱动电路

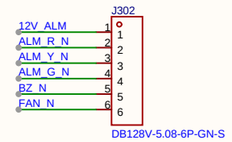


热敏打印机高边受控软启动负载开关电路

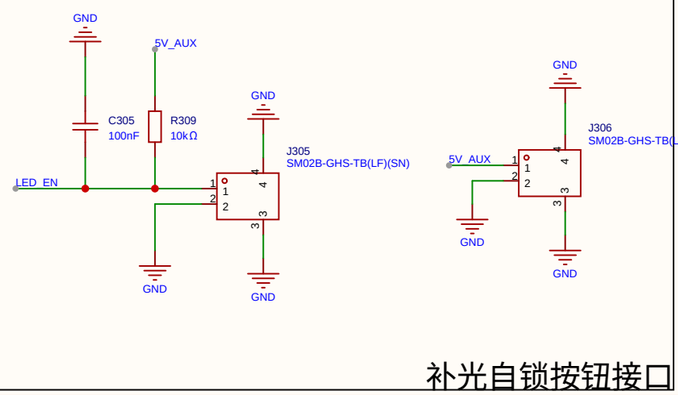


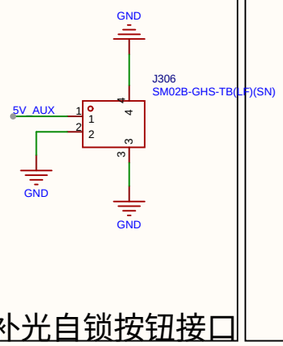
三色LED报警灯驱动电路

蜂鸣器与风扇驱动电路

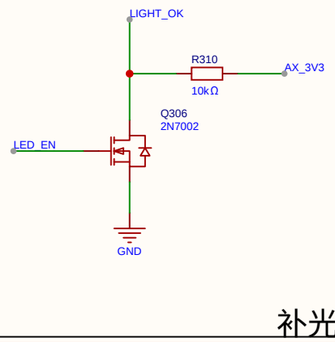


蜂鸣器、警报灯、风扇接口



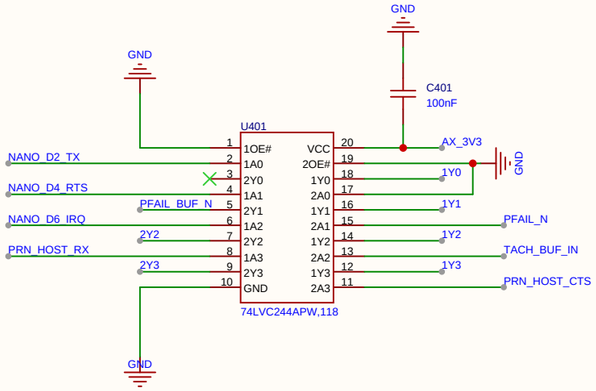


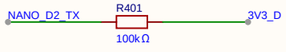
补光自锁按钮接口

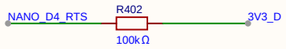


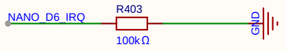
补光灯状态切换

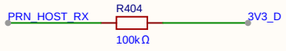
原理图	主载板原理图	创建日期	2026-10-14
		更新日期	2026-10-17
板子	主载板	图页	补光与负载驱动
绘制		FPGA_2026-10-14_17-44-11	
审阅			
		版本	尺寸
嘉立创EDA		V1.0	A4
		页 3 共 8	
		嘉立创EDA	

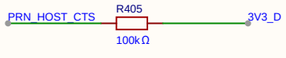






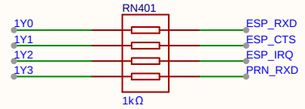


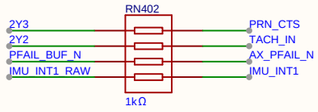




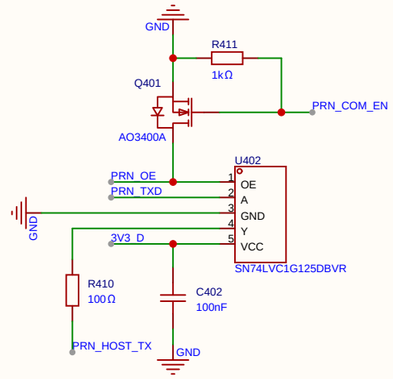


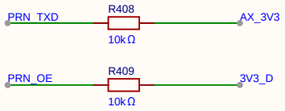


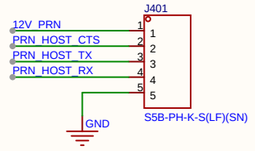




跨域连接与缓冲电路

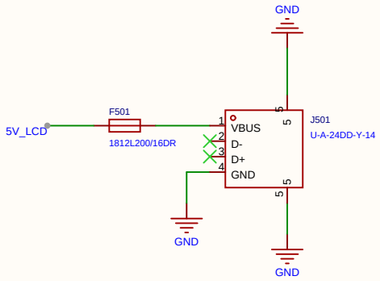




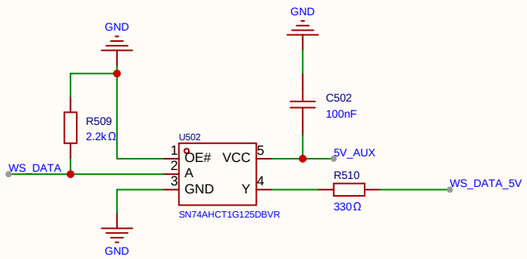


打印机通讯与接口

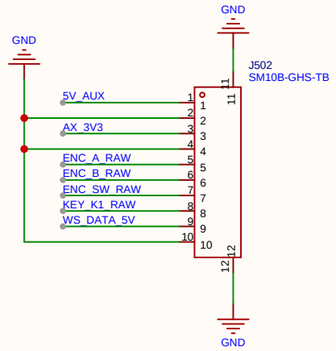
原理图	主载板原理图	创建日期	2026-10-14
		更新日期	2026-10-19
板子	主载板	图页	跨域接口与打印通信
绘制		FPGA_2026-10-14_17-44-11	
审阅			
		版本	尺寸	页 4 共 8
嘉立创EDA		V1.0	A4	嘉立创EDA

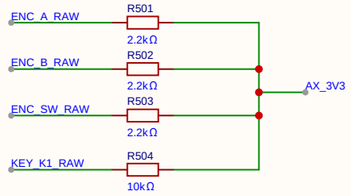


屏幕供电接口

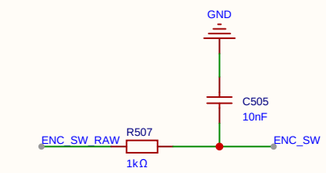


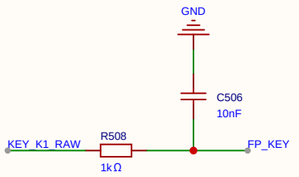
WS2812 状态灯电平转换器

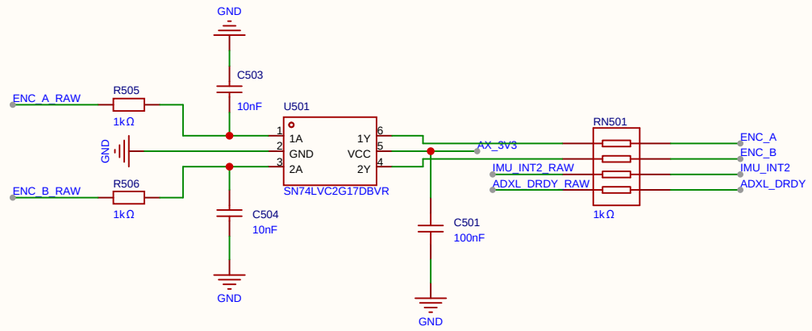




前面板 10P 接口与 4 路弱上拉

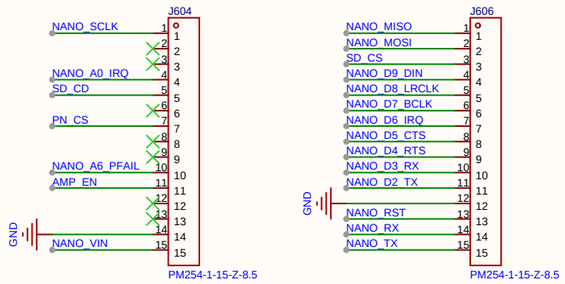


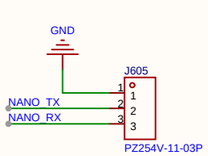


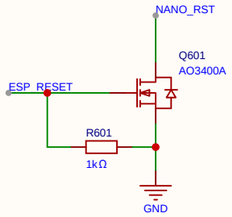


按键旋钮A/B滤波与施密特整形

原理图	主载板原理图	创建日期	2026-10-14
		更新日期	2026-10-18
板子	主载板	图页	屏幕与前面板接口
绘制		FPGA_2026-10-14_17-44-11	
审阅			
		版本	尺寸
嘉立创EDA		V1.0	A4
		页 5 共 8	
		嘉立创EDA	



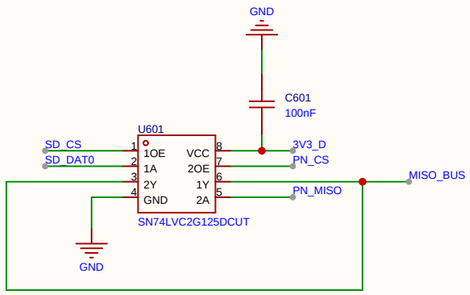


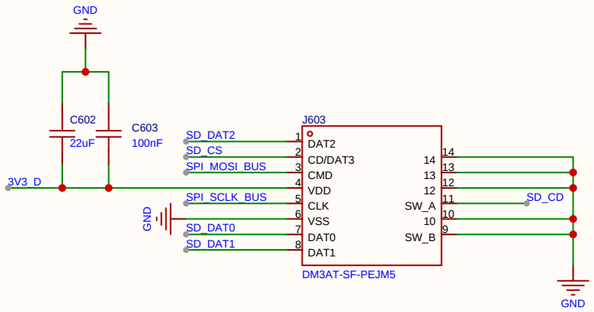


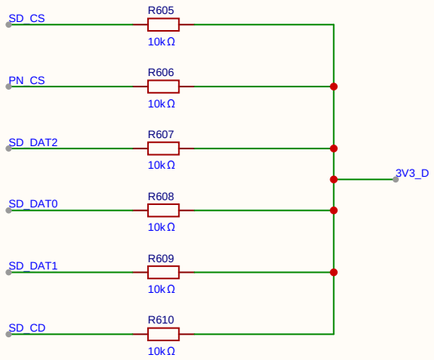




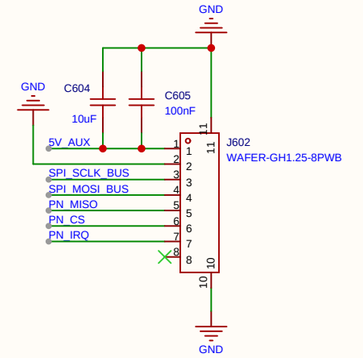


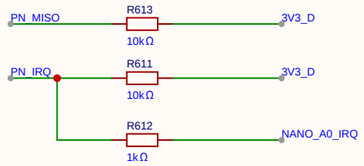




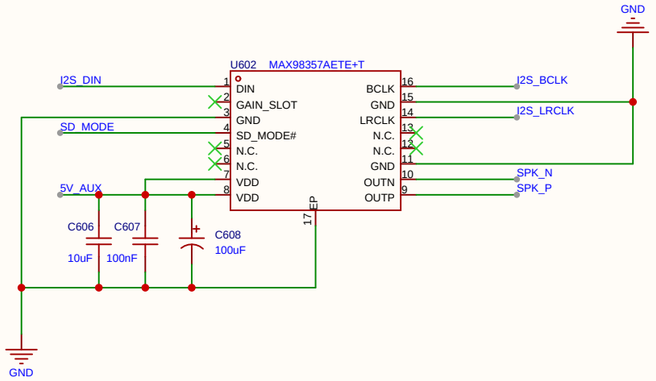


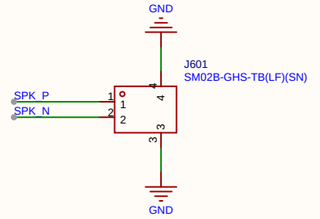
Nano与MircoSD接口

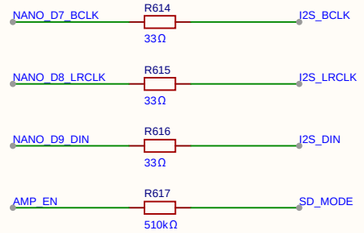




PN523接口

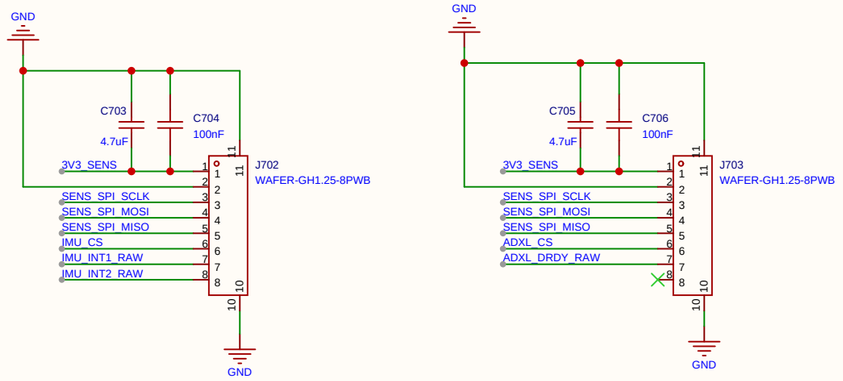


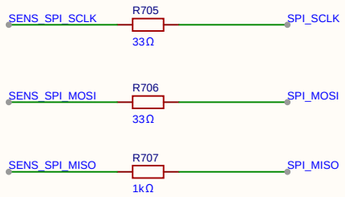




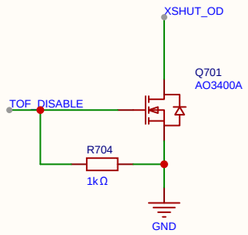
语音功放芯片与扬声器座

原理图	主载板原理图	创建日期	2026-10-14
		更新日期	2026-10-18
板子	主载板	图页	Nano 与辅助外设
绘制			
审阅			
	FPGA_2026-10-14_17-44-11		
		版本	尺寸
		V1.0	A4
嘉立创EDA		页 6 共 8	
		嘉立创EDA	

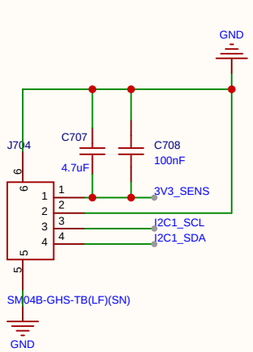


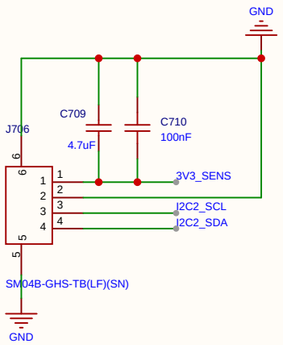


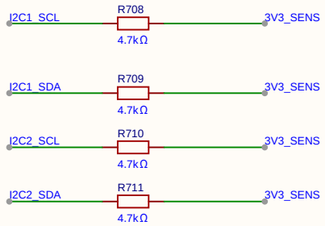
IMU接口



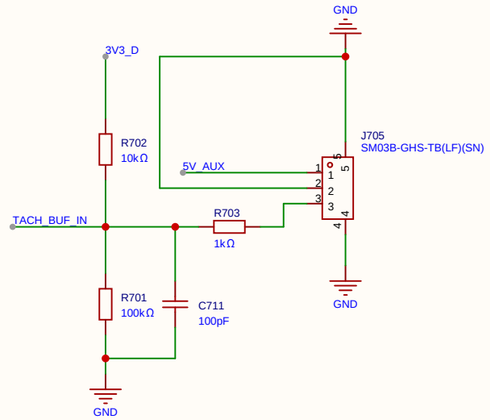
ToF接口





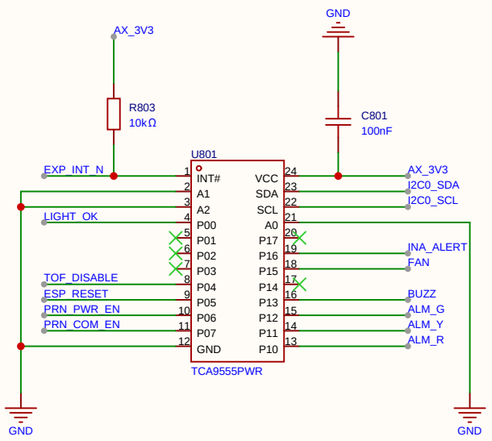


SHT40接口

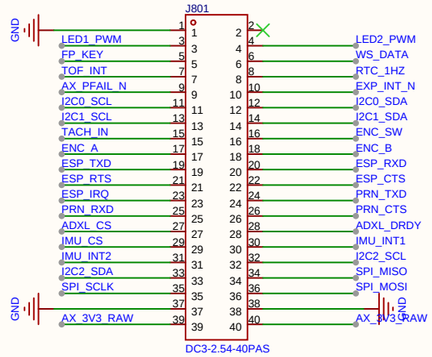


转速计扩展接口

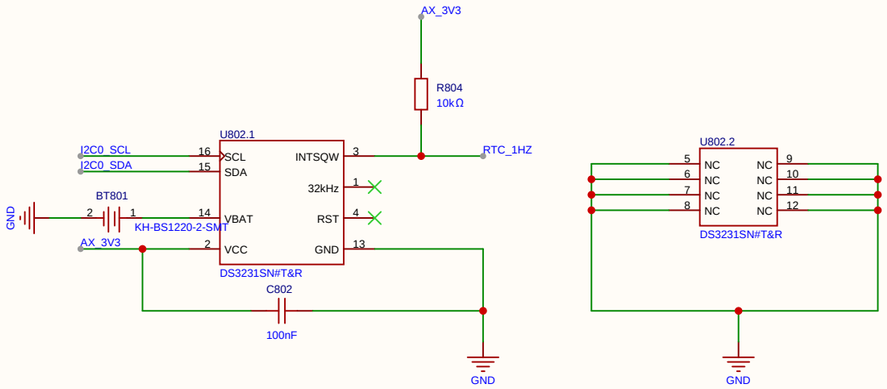
原理图	主载板原理图	创建日期	2026-10-14
		更新日期	2026-10-18
板子	主载板	图页	传感器与扩展接口
绘制		FPGA_2026-10-14_17-44-11	
审阅			
		版本	尺寸	页 7 共 8
嘉立创EDA		V1.0	A4	嘉立创EDA



慢速 IO 扩展电路



AX7010 J11 扩展牛角座接口



高精度实时时钟电路

原理图	主载板原理图	创建日期	2026-10-14
板子	主载板	更新日期	2026-10-17
绘制		图页	板内管理与主控连接
审阅		FPGA_2026-10-14_17-44-11	
		版本	尺寸
嘉立创EDA		V1.0	A4
		页 8 共 8	
		嘉立创EDA	